

Deliverable 2 – Business Requirement Document (BRD)

Healthcare Patient Experience Analytics using Power BI

Document Information

Field	Details
Project Name	Healthcare Patient Experience Analytics
Organization	National Healthcare Quality Commission (NHQC)
Department	Patient Experience & Quality Analytics
Role	Healthcare Business Intelligence Analyst
Tool Used	Microsoft Power BI
Document Type	Business Requirement Document (BRD)

1. Business Requirement

Overview

The National Healthcare Quality Commission (NHQC) requires a centralized Business Intelligence solution to analyze patient experience data collected through the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) survey.

The solution should enable healthcare executives and administrators to monitor patient satisfaction across states, regions, reporting periods, and quality measures. It should also provide interactive dashboards that help identify performance trends, compare healthcare outcomes, benchmark state performance against national averages, and support evidence-based healthcare policy decisions.

The dashboard should transform complex healthcare survey data into meaningful insights that improve decision-making and enhance the quality of patient care nationwide.

2. Functional Requirements

The Power BI solution should provide the following functionalities:

- Import and integrate multiple healthcare datasets into Power BI.
 - Create a Star Schema data model for efficient reporting.
 - Develop DAX measures for healthcare KPI calculations.
 - Display patient satisfaction metrics at national, regional, and state levels.
 - Compare state performance against national averages.
 - Analyze healthcare quality measures across reporting periods.
 - Rank states, regions, and quality measures based on performance.
 - Enable interactive filtering using slicers.
 - Provide drill-down capabilities for detailed analysis.
 - Generate actionable business insights through interactive dashboards.
-

3. Dashboard Requirements

The Business Intelligence solution should include six interactive dashboards designed for different business users.

Report 1 – Executive Overview

Audience: CEO & Executive Committee

Dashboard Requirements

- Overall Satisfaction
 - Average Top-box Percentage
 - Average Survey Response Rate
 - Number of Participating States
 - Reporting Period Analysis
 - Executive KPI Summary
-

Report 2 – State Performance Dashboard

Audience: State Health Administrators

Dashboard Requirements

- State Ranking
- State Satisfaction Score

- Top-box Percentage
 - Bottom-box Percentage
 - Regional Comparison
 - Interactive U.S. Map
 - State Performance Trend
-

Report 3 – Patient Experience Measures Dashboard

Audience: Chief Quality Officer

Dashboard Requirements

- Communication with Nurses
 - Communication with Doctors
 - Responsiveness of Hospital Staff
 - Pain Management
 - Communication about Medicines
 - Hospital Cleanliness
 - Hospital Quietness
 - Discharge Information
 - Overall Hospital Rating
 - Hospital Recommendation
 - Measure Ranking
-

Report 4 – National vs State Benchmark Dashboard

Audience: Hospital Accreditation Team

Dashboard Requirements

- National Average
 - State Average
 - Variance Analysis
 - Benchmark Comparison
 - Gap Analysis
-

Report 5 – Regional Performance Dashboard

Audience: Regional Managers

Dashboard Requirements

- Region Ranking
- Average Regional Satisfaction

- Survey Response Rate
 - Regional Trends
 - Top Performing States
 - Bottom Performing States
-

Report 6 – Trend Analysis Dashboard

Audience: Policy Makers

Dashboard Requirements

- Satisfaction Trends
- Reporting Period Comparison
- Measure Performance Trends
- Response Rate Trends
- Improvement Analysis

These dashboard requirements are based on the reporting requirements specified in the project brief.

4. User Stories

User Story 1

User: Chief Executive Officer (CEO)

Requirement:

As a **Chief Executive Officer**, I want to view a national overview of patient satisfaction so that I can monitor overall healthcare performance and support strategic decision-making.

User Story 2

User: Chief Quality Officer

Requirement:

As a **Chief Quality Officer**, I want to analyze the performance of healthcare quality measures so that I can identify areas requiring quality improvement.

User Story 3

User: Director of Patient Experience

Requirement:

As a **Director of Patient Experience**, I want to monitor patient satisfaction trends across reporting periods so that I can improve patient experience initiatives.

User Story 4

User: State Health Administrator

Requirement:

As a **State Health Administrator**, I want to compare my state's performance with regional and national averages so that I can identify opportunities for improvement.

User Story 5

User: Hospital Accreditation Team

Requirement:

As a member of the **Hospital Accreditation Team**, I want to benchmark hospital performance against national standards so that I can evaluate healthcare quality and accreditation readiness.

User Story 6

User: Policy Maker

Requirement:

As a **Policy Maker**, I want to analyze long-term patient satisfaction trends so that I can formulate effective healthcare policies and quality improvement strategies.

User Story 7

User: Healthcare Analytics Team

Requirement:

As a **Healthcare Business Intelligence Analyst**, I want to interact with healthcare data through dynamic dashboards so that I can generate actionable insights and support evidence-based decision-making.

The user stories align with the stakeholder requirements and reporting objectives described in the project brief.

5. Success Criteria

The project will be considered successful if it achieves the following objectives:

- Provides an enterprise-level interactive Business Intelligence solution.
 - Enables monitoring of patient satisfaction across national, regional, and state levels.
 - Supports KPI monitoring using DAX measures.
 - Allows users to compare healthcare performance across different dimensions.
 - Improves decision-making through interactive dashboards and visual analytics.
 - Helps identify opportunities for healthcare quality improvement.
 - Supports evidence-based healthcare planning and policy development.
-

Conclusion

The proposed Business Intelligence solution will provide a centralized platform for monitoring patient satisfaction, evaluating healthcare quality, benchmarking state and regional performance, and identifying improvement opportunities. Through interactive dashboards, KPI monitoring, and advanced analytics, the solution will support informed decision-making and contribute to continuous improvement in healthcare quality and patient experience.

